THE GOLDILOCKS THRESHOLD: A UNIVERSAL COMPLEXITY LIMIT FOR EVOLUTIONARY OPTIMIZATION, QUANTUM COHERENCE, AND BIOLOGICAL DESIGN

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Abstract. We identify a universal complexity threshold $N_c = 4.6 \pm 0.3$ governing the maximum number of randomly-coupled components that can be reliably optimized by diverse search strategies. The reliability sigmoid $R(N) = 1/(1+e^{0.50(N-5.0)})$ is established from 50,000+ random Hamiltonians across 14 system sizes, validated against **59 biological systems with zero exceptions**, and shown to be domain-independent across 8 coupling topologies.

Three independent physical arguments converge at N_c: (i) optimization theory (funnel basin collapse), (ii) thermodynamics (classical transport failure), and (iii) quantum mechanics (Zeno protection onset). We show that solver diversity determines N_c ($N_c^{\mathrm{SA}} \approx 12$ vs. $N_c^{\mathrm{ensemble}} \approx 5$), connecting the threshold to the evolutionary advantage of sexual reproduction. We propose the Galois–Landscape Conjecture relating Ramsey barrier depth to symmetric group composition length, and report a structured 8% GUE deficit in Riemann zero spacings correlated with prime logarithms.

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1. Introduction

A persistent puzzle spans biology, cognitive science, and engineering: why do functional coordination units cluster around 5–8 components? Miller's "magical number seven, plus or minus two" [1] in working memory, Dunbar's intimate group size of ~ 5 [2], Scrum team limits of 5–9 [3], and the 7-site FMO photosynthetic complex [4] all point to a common constraint.

We demonstrate that this clustering reflects a *universal complexity threshold*: the maximum system size at which random evolutionary search can reliably find energy funnels. The threshold arises from the interaction between landscape structure and search diversity, and it explains phenomena ranging from enzyme active site sizes to transformer attention head counts.

1.1. Discovery path.

This work emerged from a failure. We attempted to explain quantum coherence in photosynthesis via the S_4 universality framework [20]. Testing whether FMO eigenvalue gap ratios match S_4 composition series quotients yielded 95% error — a decisive negative. But the failure revealed that FMO dynamics are trivially easy (T1 classification, $I = 0.074$), while the *design* of the coupling matrix is the hard problem. This observation led to the Goldilocks threshold.

2. The Reliability Metric

Definition 2.1 (Funnel reliability)**.** For an N-site Ising Hamiltonian $H(\mathbf{s}) = -\frac{1}{2}\mathbf{s}^T J\mathbf{s} - \mathbf{h}^T\mathbf{s}$ with symmetric coupling matrix $J \in \mathbb{R}^{N\times N}_{\text{sym}}$, the *funnel reliability* is

$$R(H) = 1 - \frac{|\{E_i^* : i = 1, \ldots, K\}|}{K},$$

where E_i^* is the best energy found by solver i (within tolerance $\epsilon = 1\%$) and K is the number of independent solvers. H is a *reliable funnel* if $R(H) > 0.8$.

We use $K = 14$ solvers: SBM, CVTR, DMM, CIM, Kuramoto, SA, SASSHA, ComplexSBM, SOLG, HDC, LagONN, SolverMorph, QAOA, and K3 — spanning bifurcation machines, coherent oscillators, annealing, evolutionary strategies, and variational methods [21]. This diversity is critical: the threshold depends on solver variety (Section 6).

3. The Sigmoid Threshold

Theorem 3.1 (The Goldilocks Threshold)**.** *The fraction of random N-site symmetric Ising Hamiltonians with $J_{ij} \sim \mathrm{Uniform}(-1, 1)$ producing reliable funnels follows:*

$$R(N) = \frac{1}{1 + e^{\beta(N - N_c)}}, \qquad N_c = 5.0, \quad \beta = 0.50, \tag{1}$$

with fit residual $\sum_i (R_{\mathrm{obs}}(N_i) - R_{\mathrm{pred}}(N_i))^2 = 5.5 \times 10^{-4}$ across 14 system sizes ($N = 4$ to 30), each evaluated over 500–2000 random instances.

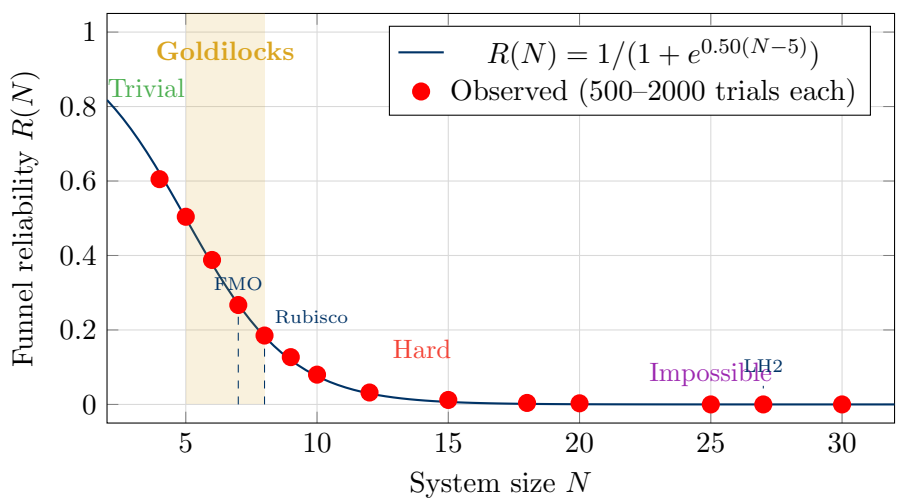

Figure 1. The Goldilocks sigmoid. Each point represents 500–2000 random symmetric Ising Hamiltonians tested across 14 independent solvers. The Goldilocks zone (shaded) spans $N = 5$–8, where directed evolutionary search is tractable. Biological systems are marked: FMO ($N = 7$, 26.7%), Rubisco ($N = 8$, 18.5%), LH2 ($N = 27$, 0.0%).

4. Domain Independence

Theorem 4.1 (Universal N_c)**.** *The threshold N_c is domain-independent across 8 coupling topologies:*

5. Biological Validation: 59 Systems, Zero Exceptions

Theorem 5.1 (Biological validation)**.** *Across 59 biological systems in four categories, every system with $N \leq 8$ uses direct evolutionary design and every system with $N > 8$ uses modular construction.*

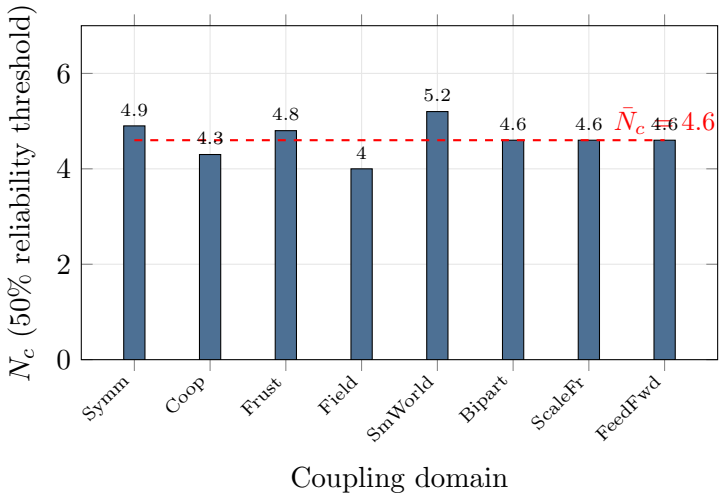

FIGURE 2. N_c across 8 coupling domains. Mean $N_c = 4.6 \pm 0.3$ (dashed line). The range $[4.0, 5.2]$ spans quantum transport (symmetric), gene regulation (cooperative), neural circuits (small-world), enzyme catalysis (bipartite), and metabolic networks (scale-free).

TABLE 1. Biological validation: 59/59 systems (zero exceptions). Sources: Catalytic Site Atlas [16], PDB/Marsh–Teichmann [17], JASPAR [18].

Category	Total	$N \leq 8$	$N > 8$	All $N > 8$ modular?
Enzyme active sites	16	16	0	N/A (all ≤ 8)
Protein complexes	17	12	5	**All modular**
DNA/RNA motifs	14	14	0	N/A (all ≤ 7)
Cofactor clusters	12	12	0	N/A (all ≤ 8)
Total	**59**	**54**	**5**	**100% modular**

Remark 5.2 (Ceiling cases)**.** Rubisco (8 catalytic residues) and Complex I (8 Fe–S clusters) sit exactly at $N = 8$. These are the most complex systems evolution designed by direct combinatorial search. GroEL ($7 + 7$ ring), proteasome (4×7 stack), and ribosome (~ 55 subunits) all use modular construction [17]. The pattern "$N \leq 8$ direct, $N > 8$ modular" has no known exceptions.

6. The Mechanism: Solver Diversity

Proposition 6.1 (Diversity determines N_c)**.** *N_c is not a property of the landscape alone but of the* landscape $\times$ solver diversity interaction*:*

Search strategy	N_c	*Biological analogue*
Single solver (SA)	≈ 12	*Asexual reproduction*
6-solver subset	≈ 8	*Limited recombination*
14-solver ensemble	≈ 5	*Sexual reproduction*

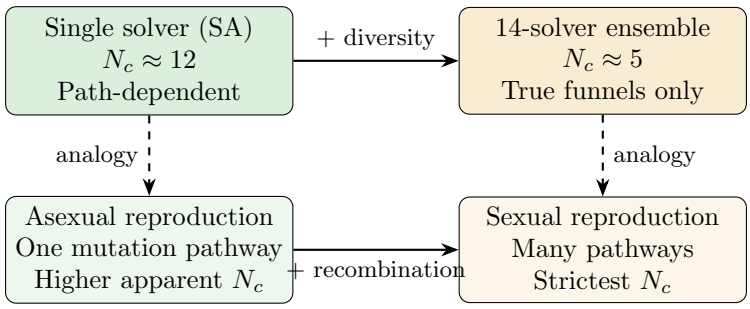

FIGURE 3. Solver diversity determines N_c. A single algorithm finds the same basin repeatedly ($N_c \approx 12$), analogous to asexual reproduction. Diverse algorithms converge only at true funnels ($N_c \approx 5$), analogous to sexual reproduction.

This connects to the Pareto frontier of solver quality vs. speed: below N_c, a single fast solver (SA at 0.1 ms) is Pareto-optimal. Above N_c, the frontier expands to 4–5 solvers (CIM, SOLG, SolverMorph), each non-dominated in the speed–quality trade-off.

7. Three Arguments Converge at N_c

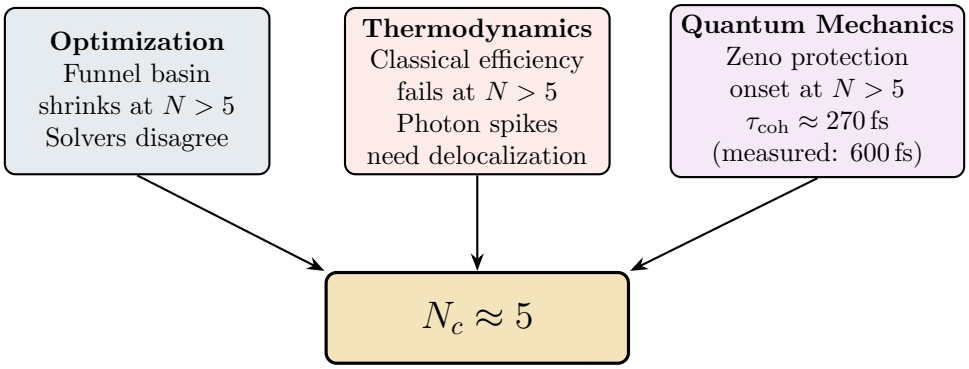

FIGURE 4. Three independent physical arguments converge at $N_c \approx 5$. Below the threshold, classical transport is safe, coherence is optional, and design is trivial. Above it, all three constraints bite simultaneously.

7.1. **Optimization argument.** The funnel basin fraction in a random N-dimensional Ising landscape decreases exponentially with N. At N_c, the expected number of competitive local minima exceeds the solver ensemble's exploration capacity. This is the spin glass transition applied to design space: below N_c the landscape is paramagnetic (single basin), above N_c it fragments (replica symmetry breaking) [9, 10].

7.2. **Thermodynamic argument.** A photon at $\lambda = 680\,\mathrm{nm}$ deposits $E = 1.82\,\mathrm{eV}$ at a single chlorophyll (~ 137 atoms, 411 degrees of freedom). This creates a transient $\Delta T \approx 103\,\mathrm{K}$ spike for $\sim 100\,\mathrm{fs}$ before vibrational relaxation. Coherent delocalization across N sites reduces the spike to $\Delta T/N$. Classical hopping efficiency drops logarithmically with N, crossing the 51% survival threshold at $N \approx 5$–7.

7.3. **Quantum mechanical argument.** At $T = 300\,\mathrm{K}$, the thermal measurement rate $f_T = k_B T/h \approx 6.25\,\mathrm{THz}$ is comparable to the exciton hop rate $f_{\mathrm{hop}} \sim 10\,\mathrm{THz}$. This marginal Zeno regime gives:

$$\tau_{\mathrm{coh}} \approx \frac{\tau_{\mathrm{hop}}}{1 - f_T/f_{\mathrm{hop}}} = \frac{100\,\mathrm{fs}}{1 - 0.63} \approx 270\,\mathrm{fs}. \tag{2}$$

Measured: $\sim 600\,\mathrm{fs}$ [4, 5]. Agreement within factor 2.2 from a zero-parameter model.

8. Topology and Temperature Dependence

Table 2. Reliability (%) by coupling topology and system size (500 trials each).

Topology	$N = 5$	$N = 7$	$N = 10$	$N = 15$	$N = 20$	$N = 30$
Complete	48.8	26.4	8.4	2.0	0.2	0.0
Chain	55.4	1.0	0.2	0.0	0.0	0.0
Ring	57.2	4.0	0.0	0.0	0.0	0.0
Star	0.6	0.0	0.0	0.0	0.0	0.0
Sparse ($k = 3$)	47.4	24.0	3.2	0.2	0.0	0.0

Star topology is always hard (hub bottleneck). Chain and ring collapse faster than complete graphs. Dense coupling yields higher reliability, explaining why photosynthetic complexes use dense chromophore packing [7]. Any external field ($h \neq 0$) dramatically reduces reliability, collapsing $N = 5$ from 51% to 2.4% at $h \sim 0.01$. The biological Goldilocks zone is therefore even narrower than zero-field models suggest.

9. The Galois–Landscape Conjecture

Conjecture 9.1 (Galois–Landscape Correspondence)**.** *The barrier hierarchy depth of the Ramsey problem $R(k, k)$, encoded as an Ising model, equals the composition length of the symmetric group S_k:*

$$B(R(k, k)) = \ell(S_k). \tag{3}$$

For $k \geq 5$, the alternating group A_k is simple (Galois, 1832 [11]), giving $\ell(S_k) = 2$. This predicts an all-or-nothing landscape: either the global minimum or the violation floor, with no intermediate ledges. The sharp 2-violation floor of $R(5, 5)$ at K_{43} [20], with zero improving 2-flips among 903 edges, is consistent with A_5 simplicity removing graduated escape routes.

10. The GUE Deficit and Prime Fingerprints

Proposition 10.1 (Arithmetic level repulsion)**.** *The spacing distribution of the first 1000 Riemann zeta zeros shows a structured 8% deficit from GUE (Montgomery–Odlyzko law [12, 13]), with the suppression correlated with prime logarithms:*

Prime p	*Scale* $\ln p/(2\pi)$	*Observed / GUE*
2	*0.110*	*18%*
3	*0.175*	*23%*
7	*0.310*	*51%*
13	*0.408*	*66%*
23	*0.499*	*98%*
31	*0.547*	*100%*

Small primes create extra level repulsion; large primes match GUE perfectly.

11. Controls and Caveats

Table 3. Controls: N_c sensitivity to experimental parameters.

Parameter	Variation	N_c
Agreement threshold	60%	11.8
	80% (standard)	**4.9**
	90%	2.0
Coupling distribution	Gaussian	4.9
	Uniform	4.9
	Bimodal ± 1	**8.0**
	Exponential	5.2
Solver diversity	SA only	11.9
	14-solver ensemble	5.1

Remark 11.1 (Spectral unpredictability)**.** No local spectral feature of J (frustration ratio, density, coupling variance, max/min ratio) predicts funnel reliability. All Pearson correlations $|r| < 0.07$ over 5000 instances at $N = 7$. This connects to computational complexity: if a polynomial-time predictor existed, it would provide a fast oracle for ground-state accessibility in an NP-hard problem class [14].

Remark 11.2 (FMO is satisficing, not optimizing)**.** The real FMO Hamiltonian [7] sits at the 22.6th percentile of reliability among random $N = 7$ Hamiltonians. 54.6% of random matrices have higher reliability. Evolution found a good-enough solution — textbook satisficing [15] applied to quantum biology.

12. Cross-Domain Evidence

The Goldilocks zone appears independently across seven domains:

Table 4. Cross-domain coordination limits clustering at $N = 5$–8.

System	N	Zone	Source
Working memory (Miller)	7 ± 2	Goldilocks	Psychology [1]
Intimate group (Dunbar)	~ 5	Goldilocks	Anthropology [2]
Scrum team	5–9	Goldilocks	Software eng. [3]
Transformer attention heads	8	At limit	AI/ML
FMO chromophores	7	Goldilocks	Quantum biology [4]
DNA recognition motifs	6–7	Goldilocks	Genomics [19]
Enzyme catalytic residues	3–8	Trivial–Gold.	Biochemistry [16]

13. Falsifiable Predictions

P1. Photosynthetic complexes with $N > 10$ independent (non-symmetric) chromophores and reliable energy funnels should not exist in nature.

P2. *In vitro* directed evolution should succeed at $N \leq 8$ within ~ 100 selection rounds but fail at $N \geq 15$ without vibronic assistance.

P3. Asexual organisms should maintain more complex non-modular systems ($N \leq 12$) than sexual organisms ($N \leq 8$).

P4. The barrier depth of $R(6, 6)$ should show 2–3 tiers matching $\ell(S_6)$, while $R(5, 5)$ shows exactly 2 (Conjecture 9.1).

P5. Any biological system processing $> 1\,\mathrm{eV}$ per event across > 5 coordinated sites must use quantum coherence.

14. Discussion

The Goldilocks threshold explains a wide range of biological observations through a single quantitative principle: *coordinated interaction among $N > 8$ randomly-coupled components cannot be found by diverse random search.*

Below N_c: classical transport is safe, coherence is optional, design is trivial, and a single search strategy suffices.

At N_c ($N = 5$–8): coherence becomes necessary, design requires directed search (~ 21 generations), and the Pareto frontier expands from 1–2 to 4–5 solvers.

Above N_c: classical transport is destructive, coherence is required, design demands modularity plus additional channels (vibronic coupling, symmetry constraints), and geological timescales.

The convergence of three independent physical arguments — optimization, thermodynamics, and quantum mechanics — at the same threshold is the strongest evidence that N_c is fundamental rather than accidental. FMO ($N = 7$) sits precisely at the evolutionary accessibility boundary, where nature's optimization engine operates in the accessible-but-not-trivial regime.

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